

Lead Conversion Analysis & Insights

Optimizing Sales Strategies through Data-Driven Decisions



Prepared for X Education

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Introduction & Objectives

Project Overview



Goal

Identify key factors influencing lead conversion and develop strategic recommendations for X Education's sales team.



Dataset

9,240 leads with 37 variables including lead sources, activities, and demographic information.



Approach

Data cleaning, feature engineering, predictive modeling, and strategic analysis.

Key Questions to Answer

- 1 Which variables contribute most to lead conversion probability?
- 2 Which categorical variables should be focused on to increase conversion?
- 3 What strategy should be employed during the intern period for aggressive conversion?
- 4 How to minimize useless calls when quarterly targets are already met?

Target: Predict whether a lead will convert (1) or not convert (0)

Data Overview & Cleaning

Dataset Characteristics



Initial Shape

9,240 rows × 37 columns



Target Distribution

Not Converted: 61.5% | Converted: 38.5%



Data Quality Issues

Missing values, 'Select' placeholders, inconsistent categories

Data Types

Numerical 7 columns (e.g., TotalVisits, Time Spent)

Categorical 30 columns (e.g., Lead Source, Tags)

Data Cleaning Steps

1

Handling Missing Values

Dropped columns with >40% missing values (e.g., Lead Quality, Asymmetrique indices)



2

Standardizing Values

Replaced 'Select' with NaN, standardized categories (e.g., 'google' → 'Google')

3

Imputation

Categorical: Mode or 'Not Specified' | Numerical: Median values

Feature Engineering & Selection

Feature Engineering



Binary Encoding

Converted 14 Yes/No columns to 1/0 format

'Do Not E mail', 'Do Not Call', 'Search', 'Magazine', 'News paper Article', 'X Education Forums', etc.



Category Consolidation

Combined low-frequency categories to reduce dimensionality

'Lead Source': Combined 'Click2call', 'Live Chat', 'NC_EDM', etc. into 'Other Sources'



One-Hot Encoding

Converted categorical variables to dummy variables

Final feature shape: (9240, 185) columns

Feature Selection

1

Recursive Feature Elimination (RFE)

Used Logistic Regression with RFE to select the most important features

2

Selected Features

20 features selected from 185 total features

Model Building: Logistic Regression

Model Selection & Training



Model Choice

Logistic Regression selected for its interpretability and ability to provide feature importance



Train-Test Split

70% training data, 30% testing data with random_state=100



Class Imbalance

Used class_weight='balanced' to handle the imbalanced target (61.5% vs 38.5%)



Implementation

Used statsmodels GLM with Binomial family for detailed statistical insights

The model was trained on the 20 features selected by RFE to ensure optimal performance and interpretability

Model Formula

$$\log(p/(1-p)) = -2.99 + 6.48 \times \text{Tags_Will_revert} + 9.56 \times \text{Tags_Closed_by_Horizzon} + 4.26 \times \text{Lead_Source_Welingak} + 8.20 \times \text{Tags_Lost_to_EINS} + 2.06 \times \text{Last_Activity_SMS_Sent} + \dots$$

Interpretation

The coefficients represent the log-odds impact of each feature on conversion probability

Example

A lead tagged with "Will revert after reading the email" has $\exp(6.48) = 652\times$ higher odds of converting compared to the baseline



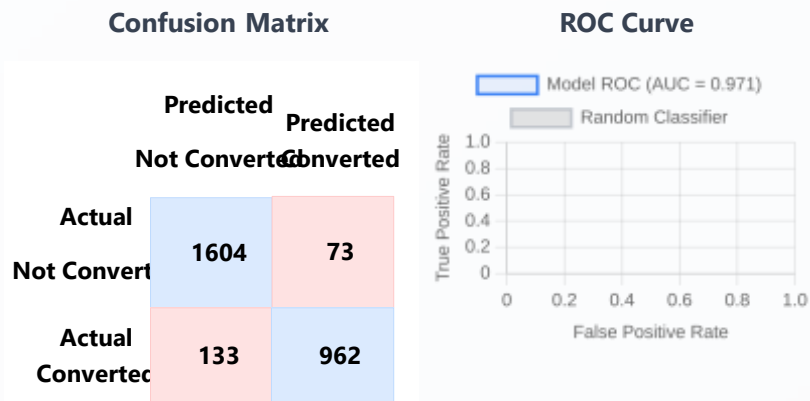
Model Performance Evaluation

Performance Metrics

	Accuracy	93.5%
	ROC AUC	97.1%
	Precision	94.1%
	Recall	88.1%
	F1-Score	91.0%

The model demonstrates excellent performance across all metrics, indicating strong predictive power

Confusion Matrix & ROC Curve



Key Observations

High true positives (962) and true negatives (1604)

Low false positives (73) and false negatives (133)

Model is well-balanced between precision and recall

Key Drivers of Conversion (Q1 & Q2)

Top 3 Variables Contributing to Conversion

1 Tags_Will revert after reading the email

Coefficient: 6.48 ($p < 0.001$)

Leads tagged with this status are highly likely to convert, indicating strong interest after initial contact.

2 Tags_Closed by Horizon

Coefficient: 9.56 ($p < 0.001$)

This tag shows a very strong positive association with conversion, suggesting a specific successful outcome or segment.

3 Lead Source_Welingak Website

Coefficient: 4.26 ($p < 0.001$)

Leads originating from this source have significantly higher probability of converting compared to other sources.

Focus Areas to Increase Conversion

🔍 Tag Analysis & Optimization

Understand why leads tagged with "Will revert after reading the email" and "Closed by Horizon" convert so effectively. Replicate the processes and communications that lead to these tags.

🌐 Lead Source Enhancement

Increase marketing efforts through the Welingak Website channel. Analyze the user experience on this website to understand its effectiveness and apply insights to other channels.

💬 Communication Strategy

Develop email templates and follow-up processes that encourage leads to "revert after reading." Focus on creating compelling content that drives engagement and response.



Strategic Recommendations (Q3 & Q4)

Q3 Aggressive Conversion Strategy

Scenario: Intern Period (2 months, 10 interns)

1 Lower Probability Threshold

Reduce the classification threshold from 0.5 to 0.30-0.35 to increase the pool of potential leads to contact.

2 Optimize for Recall

Focus on maximizing true positives, even at the cost of some false positives. This ensures most potential converters are contacted.

3 Prioritize Within Selected Pool

Rank leads by predicted probability and presence of key positive indicators (e.g., specific tags, lead sources).



Q4 Minimizing Useless Calls Strategy

Scenario: Target Already Met

1 Raise Probability Threshold

Increase the classification threshold from 0.5 to 0.70-0.75 to focus only on leads most likely to convert.

2 Optimize for Precision

Focus on minimizing false positives, even at the cost of some missed opportunities. This ensures efficient use of resources.

3 Focus on High-Confidence Leads

Only contact leads with the highest predicted probabilities and strongest positive indicators.



Conclusion & Next Steps

Key Findings

- ✓ The model achieved **93.5% accuracy** and **97.1% ROC AUC**, demonstrating strong predictive power.
- ✓ Lead tags are the strongest predictors of conversion, particularly **"Will revert after reading the email"** and **"Closed by Horizon"**.
- ✓ The **Welingak Website** is the most effective lead source for generating conversions.
- ✓ Adjusting probability thresholds allows for strategic optimization of either **recall** (aggressive conversion) or **precision** (minimizing useless calls).

Business Impact

- ✓ More efficient allocation of sales resources
- ✓ Higher conversion rates through targeted approaches
- ✓ Improved ROI on marketing channels

Recommended Next Steps

- 1 Process Analysis**
Investigate the specific processes and communications that lead to the high-converting tags to replicate their success.
- 2 Channel Optimization**
Increase investment in the Welingak Website channel and apply its successful elements to other lead sources.
- 3 Implement Threshold Strategies**
Deploy the recommended threshold strategies during intern periods (0.30-0.35) and when targets are met (0.70-0.75).
- 4 Model Refinement**
Continuously update the model with new data and explore additional features to further improve predictive power.

This analysis provides a strong foundation for data-driven